

cones—about 150,000 cones per square millimeter. The ratio of cones to rods falls with distance from the fovea, with the peripheral regions having only rods. In daylight, you achieve your sharpest vision by looking directly at an object, such that light shines on the tightly packed cones in your fovea. At night, looking directly at a dimly lit object is ineffective, since the rods—the more sensitive light receptors—are absent from the fovea. For this reason, you see a dim star best by focusing on a point just to one side of it.

CONCEPT CHECK 50.3

1. Contrast the light-detecting organs of planarians and flies. How is each organ adaptive for the lifestyle of the animal?
2. In a condition called presbyopia, the eyes' lenses lose much of their elasticity and maintain a flat shape. How would you expect this condition to affect a person's vision?
3. **WHAT IF?** Our brain receives more action potentials when our eyes are exposed to light even though our photoreceptors release more neurotransmitter in the dark. Propose an explanation.
4. **MAKE CONNECTIONS** Compare the function of retinal in the eye with that of the pigment chlorophyll in a plant photosystem (see Concept 10.2).

For suggested answers, see Appendix A.

CONCEPT 50.4

The senses of taste and smell rely on similar sets of sensory receptors

Animals use their chemical senses for a wide range of purposes, including finding mates, recognizing marked territories, and helping navigate during migration. In addition, animals such as ants and bees that live in large social groups rely extensively on chemical “conversation.”

In all animals, chemical senses are important for feeding behavior. The perceptions of **gustation** (taste) and **olfaction** (smell) both depend on chemoreceptors. In the case of terrestrial animals, taste is the detection of chemicals called **tastants** that are present in a solution, and smell is the detection of **odorants** that are carried through the air. There is no distinction between taste and smell in aquatic animals.

In insects, taste receptors are located within sensory hairs located on the feet and in mouthparts, where they are used to select food. A tasting hair contains several chemoreceptors, each especially responsive to a particular class of tastant, such as sugar or salt. Insects are also capable of smelling airborne odorants using olfactory hairs, usually located on their antennae (see Figure 50.6). The chemical DEET (*N,N*-diethyl-meta-toluamide), marketed as an insect “repellent,” actually protects against bites by blocking the olfactory receptor in mosquitoes that detects human scent.

Taste in Mammals

Humans and other mammals perceive five tastes: sweet, sour, salty, bitter, and umami. Umami (which means “pleasant savory taste” in Japanese) is elicited by the amino acid glutamate. Sometimes used as a flavor enhancer in the form of monosodium glutamate, or MSG, glutamate occurs naturally in meat, aged cheese, and other foods, to which it imparts a “savory” quality.

For decades, many researchers assumed that a taste cell could have more than one type of receptor. An alternative idea is that each taste cell has a single receptor type, programming the cell to recognize only one of the five tastes. To test this hypothesis, scientists used a cloned bitter taste receptor to genetically reprogram gustation in a mouse (**Figure 50.23**). This reprogramming experiment, together with follow-up studies, revealed that an individual taste cell does in fact express a single receptor type and detects tastants representing only one of the five tastes.

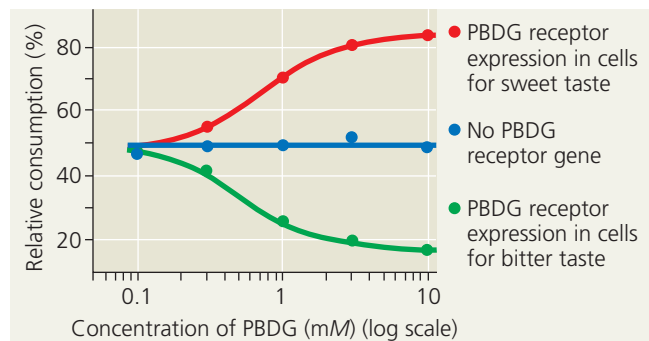
▼ Figure 50.23 Inquiry

How do mammals detect different tastes?

Experiment To investigate the basis of mammalian taste perception, researchers used a chemical called phenyl- β -D-glucopyranoside (PBDG). Humans find PBDG extremely bitter. Mice, however, lack a receptor for PBDG. Mice avoid drinking water containing other bitter tastants but show no aversion to water that contains PBDG.

Using a molecular cloning strategy, the researchers generated mice that made the human PBDG receptor in cells that normally make either a sweet receptor or a bitter receptor. The mice were given a choice of two bottles, one filled with pure water and one filled with water containing PBDG at varying concentrations. The researchers then observed whether the mice had an attraction or an aversion to PBDG.

Results



Conclusion The researchers found that the presence of a bitter receptor in sweet taste cells is sufficient to cause mice to be attracted to a bitter chemical. They concluded that the mammalian brain must therefore perceive sweet or bitter taste solely on the basis of which afferent neurons are activated.

Data from K. L. Mueller et al., The receptors and coding logic for bitter taste, *Nature* 434:225–229 (2005).

WHAT IF? Suppose that instead of the PBDG receptor, the researchers had used a receptor specific for a sweetener that humans crave but mice ignore. How would the results of the experiment have differed?